SPY Long Straddle Backtest

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Prepared for Z1 Tech

github.com/radvect/spy-long-straddle-backtest

AI Assistance & Scope

AI Assistance

ChatGPT and Claude assisted with software development and preparation of this presentation.

Scope of This Work

This project focuses on the formulation, implementation, and computational validation of a model-based options backtest.

Some financial details are simplified so that I can explain the project in mathematical terms.

All results depend on the assumptions of the model.

Long Straddle: From Contracts to a Payoff

CALL + PUT, ONE STRIKE K, ONE EXPIRY T

SPY tracks the S&P 500. The **strike** K is the fixed price in the contract. A **call** allows buying SPY at K; a **put** allows selling at K. Both expire at T, when SPY has price S_T.

Call and put values at expiration (per share)

$$C_T = \max(S_T - K, 0),$$
$$P_T = \max(K - S_T, 0).$$

$$\boxed{C_T + P_T = |S_T - K|}$$

Net profit or loss for one straddle

$$\boxed{\Pi_T = 100(|S_T - K| - C_0 - P_0) - F}$$

C_0 and P_0 are purchase prices per share; F is the total trading cost. Each contract covers 100 shares.

ILLUSTRATIVE PAYOFF

Example: $K = \$500$; purchase price plus fees $= \$12$ per share



Break-even prices: $K \pm \$12$. Maximum loss: \$12 per share.

Strategy: Four Rules

Repeat the same trade every month.

1. **Buy:** one call and one put with the same strike, closest to the current SPY price.
2. **When:** select an entry date about 30 days before a monthly expiration (allowed DTE: 21–45 days); adjust for market holidays.
3. **Exit:** hold the position until expiration and calculate its payoff.
4. **Constraint:** one straddle at a time. Skip a trade if it cannot be purchased with available cash.

Pricing and Data

The payoff is known at expiration. We must estimate the purchase price.

1. **Pricing model.** Black–Scholes estimates the call and put purchase prices from the SPY price, strike, time remaining and a volatility input. Full equations are in Appendix A.

2. **Historical prices.** We use daily SPY prices from 2020–2025 to simulate how the strategy would behave.

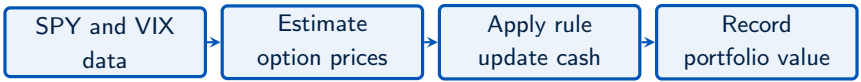
3. **Volatility estimate.** VIX (Cboe Volatility Index) summarizes the market's expectation of S&P 500 price fluctuations over approximately 30 days. We use VIX/100 as an approximate annual volatility input.

Prices of SPY and VIX are observed; prices of the options are calculated. The default annual risk-free rate is 1.0% and dividend yield is 1.3%.

Model reference: Black & Scholes (1973), *Journal of Political Economy*.

Backtest Algorithm

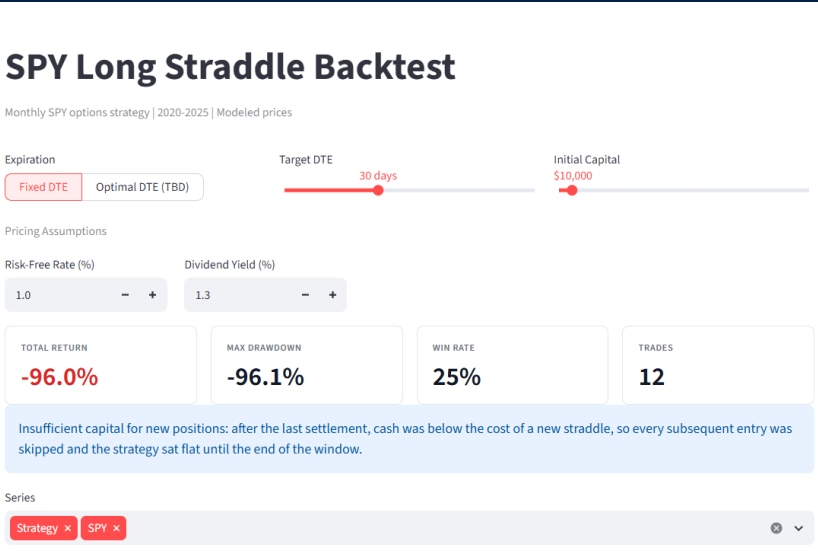
Follow the data in chronological order.



$$\text{Portfolio value} = \text{Cash} + \text{Value of any open position}$$

- At entry, record the chosen strike, expiration, purchase price and trading costs.
- At expiration, settle the position using the payoff formula from the mathematical slide.
- Record portfolio value and drawdown on every date.

Dashboard: Controls and Summary



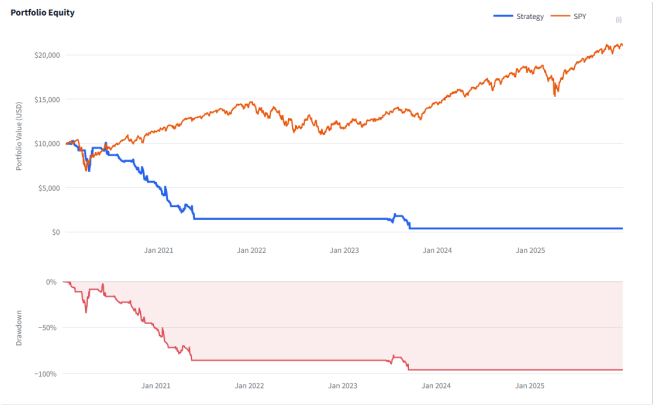
Dashboard: Trade History

Trade History

ENTRY DATE	EXPIRATION	STRIKE	DTE	ENTRY COST	EXIT VALUE	NET P&L
2020-01-22	2020-02-21	331	30	$1,006.76	$248.00	$-773.72
2020-03-18	2020-04-17	240	30	$4,278.39	$4,664.00	$+282.33
2020-05-20	2020-06-19	297	30	$1,947.06	$1,164.00	$-816.34
2020-07-22	2020-08-21	327	30	$1,863.08	$1,248.00	$-650.04
2020-09-16	2020-10-16	339	30	$2,066.80	$829.00	$-1,264.38
2020-10-21	2020-11-20	343	30	$2,299.31	$1,233.00	$-1,100.97
2020-12-16	2021-01-15	370	30	$1,950.68	$570.00	$-1,402.08
2021-01-20	2021-02-19	384	30	$1,941.13	$603.00	$-1,360.19
2021-03-17	2021-04-16	397	30	$1,789.97	$2,026.00	$+185.51
2021-04-21	2021-05-21	416	30	$1,706.94	$106.00	$-1,613.06

Backtest Results

JAN 2020–DEC 2025 | INITIAL CAPITAL $10,000 | TARGET 30 DTE



TOTAL RETURN

−96.0%

MAXIMUM DRAWDOWN

−96.1%

WIN RATE

25%

TRADES

12

What Do These Results Tell Us?

Most of the completed straddles did not recover their purchase cost.

RESULT OF 12 TRADES

Unprofitable: 9



Profitable: 3



FINAL PORTFOLIO VALUE

$402.02

The 12 recorded trade profits and losses sum to −$9,597.98 from an initial $10,000.

WHY DOES THE EQUITY CURVE FLATTEN?

Once available funds became too small for another straddle, new entries were skipped.

Validation & Testing

Check formulas first, then the complete backtest.

1. **Unit tests: mathematical functions.** Black–Scholes put–call parity and price bounds; ATM strike selection; expiration and DTE rules.
2. **Integration tests: full trade.** Entry, cost deduction, the 100-share multiplier, expiration payoff, and portfolio accounting.
3. **Scenario tests: known market paths.** Constant prices, large moves, market holidays, and insufficient funds.
4. **Regression tests: repeatable calculations.** Fixed synthetic input data, portfolio-value reconciliation, and performance statistics.

Timing check: verify that a trade never uses data that would become available only after execution.

Conclusion, Limitations and Next Steps

OBSERVED RESULT

This fixed strategy lost money

Over the recorded period, the straddle strategy performed worse than buying and holding SPY.

MODEL LIMITATIONS

- Option prices are estimated rather than observed.
- VIX is only an approximation for the volatility of each option.
- Black–Scholes omits early exercise and sudden price jumps.

Live dashboard demonstration.

FUTURE WORK

Improve the model and the experiment

- Compare modeled option prices with historical option quotes.
- Compare volatility models and test assumptions about trading costs.
- Add parameter-sensitivity plots and alternative exit rules to the dashboard.

Appendix A: Black–Scholes With Dividend Yield

$$C_t = S_t e^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),$$

$$P_t = Ke^{-r\tau}\Phi(-d_2) - S_t e^{-q\tau}\Phi(-d_1),$$

$$d_1 = \frac{\ln(S_t/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Consistency identity: $C_t - P_t = S_t e^{-q\tau} - Ke^{-r\tau}$ **(put–call parity).**

C_t, P_t: call and put prices at time t; S_t: SPY price; K: strike; τ: years to expiration; σ: annual volatility; r: interest rate; q: dividend yield; Φ: cumulative distribution function of the standard normal distribution.

This model prices European-style options. SPY options allow early exercise.

References: Black & Scholes (1973); Merton (1973), dividend extension.

Appendix B: Implementation Details

PORTFOLIO ACCOUNTING

- At purchase, subtract $100(C_0 + P_0)$ and entry costs from cash (C_0, P_0: purchase prices per share).
- While holding, the position value is $100(C_t + P_t)$, where C_t, P_t are current model prices per share.
- At expiration, add $100|S_T - K|$ (minus applicable exit costs) to cash.
- With no position, equity equals cash.

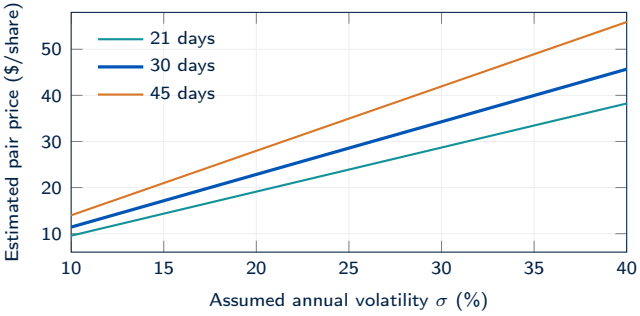
ASSUMPTIONS

- VIX estimates 30-day S&P 500 volatility. We approximate annual volatility by $\sigma_t = \mathrm{VIX}_t/100$.
- Black–Scholes computes European-option prices. SPY options permit earlier exercise.
- Commissions and assumed buying/selling spreads are included in trade costs.
- Daily closing prices do not by themselves define executable trade times; timing is an important backtest limitation.

Appendix C: Price Sensitivity

EXAMPLE: SAME $S = K = \$500$, DIFFERENT VOLATILITY AND TIME

Calculated price of one call + one put



Calculated model prices (not backtest returns); $r = 1\%$, $q = 1.3\%$.

Two simple observations

1. Greater expected price variation makes the option pair more expensive.
2. More time until expiration also increases its price in this example.